MOTOR: Musculoskeletal Optimisation Through Open Robotics Simulation

J.P. Kardolus, Babon Innovations B.V., Utrecht, The Netherlands

Working draft, 7 November 2026

Abstract

MOTOR is an open chain from one ordinary video to the ground reaction force under each foot, the joint moments and the force in each of 416 muscles, on one GPU. A monocular estimator gives a parametric body per frame; a full-body musculoskeletal model is fitted to it and stood on the floor the camera sees; a residual reduction makes the movement agree with the ground; and a two-stage static optimisation gives the ground reaction force, then the muscle activations. Every stage runs on Newton and MuJoCo Warp, deterministically. It takes a median 66 s per trial on one card, where OpenCap's recorded solves take 44 to 70 min on the CPU. On OpenCap's LabValidation cohort (715 of 720 single-camera inputs) the matched joint angle error is 5.16° (centred 3.06°). On the five subjects never used to choose a setting, the vertical force per foot follows the plates with a median correlation of 0.78 and 0.79, and the vertical ground reaction force error fell from 54 to 16% body weight (11.6% on the 45° camera; OpenCap's two cameras: 11.4%). Joint moments (2.4% weight times height, OpenCap 1.2) and the pelvis residual are not yet at the lab's level. MOTOR is open source under the MIT licence.

1 Introduction

Joint angles from video are now close to laboratory accuracy with one or two smartphones [1–4]. The force under each foot and in each muscle still comes from a slow CPU pipeline in OpenSim [5–7]; OpenCap brought it to smartphone video, at tens of minutes of solving per trial [1].

Much of what MOTOR does was shown first by others. Cotton fitted biomechanical models to video as differentiable models in MuJoCo on the GPU [8]; the Portable Biomechanics Laboratory fits one body scale jointly across all of a person's trials from a handheld phone [3]; KinTwin recovers torques, muscle activations and ground reaction forces from markerless capture [9]; and Cotton et al. fit a person's constants over frames sampled from all their recordings [10], an idea MOTOR's fit uses. MyoSuite [11] provides the muscle model, LocoMuJoCo [12] and MuJoCo [13] the simulation lineage; MuscleMimic [14] and MS-Emulator [15] run bodies of 400 to 700 muscles on MuJoCo Warp, and static optimisation on MuJoCo models has run on the CPU [16, 17].

MOTOR's own part is narrow, and its case is speed and depth: the whole chain from video to muscle forces (fit, footing, residual reduction, ground reaction force per foot, two-stage static optimisation) on MyoFullBody's 416 muscles, every stage on one GPU in about a minute per trial, bit-for-bit repeatable, and validated on a public cohort on forces, moments and muscles as well as angles. To our knowledge it is the first open chain from video to muscle forces on Newton [18] and MuJoCo Warp [19]. It is the sim body of SCORE, an open environment synthesis framework, run here on real video.

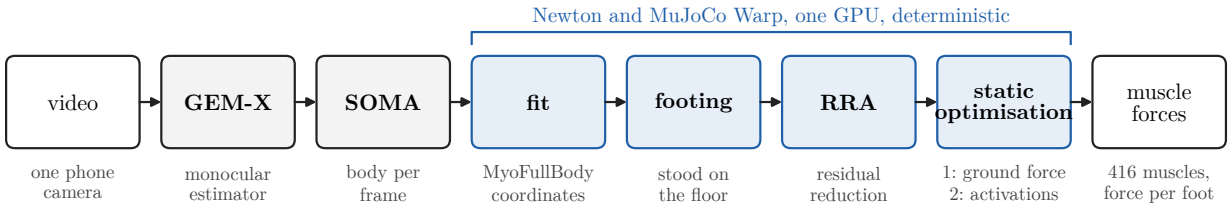


Figure 1: MOTOR's chain from one video to muscle forces. The stages in blue run every frame at once on one GPU.

2 Method

From video to a body. GEM-X [20, 21] turns the video into a SOMA body [22] per frame at 30 Hz, keeping its identity, root translation and global segment rotations. SOMA is the truth: its shape and pose are never changed. GEM-X is given the camera's field of view that MoGe-2 [23] estimates from the same video, and its own contact post-process holds the static feet. Its world has no floor, so the floor is seen instead: fitted to MoGe-2's floor points beside the person, the clip is turned onto it in one rigid turn and lowered until its feet's lowest point meets it.

Fit. The body is MyoFullBody [11] (arm and hand from MoBL-ARMS [24, 25]): 416 muscles, 86 bones and 131 joints, 95 of them free. Its bone groups are sized to the person's SOMA body and its masses spread by SOMA's segment volumes over the subject's measured mass. One solver fits the person's constants once, on frames sampled across the recordings [10], then every frame's joint angles against each SOMA segment's rotation, 84 anatomical landmarks on the SOMA skin and SOMA's joints, with the model's joint couplings held exactly and a cost that keeps the bones inside the skin. On the card it equals the CPU reference (mink [26]) within 0.01° and 0.01 mm.

Footing and ground. The standing feet are stood on the floor by moving the whole body up or down, never by changing the pose, and GEM-X's vertical drift between flights is taken off the root alone. The ground is what the camera sees: the floor, a sit-to-stand's 40 cm chair [1] and a drop jump's box, each a drawn surface. It pushes only on points of the person's SOMA skin, carried by the bones, that touch it; a foot GEM-X sees standing is pushed from its sole up to 5 cm. That height, the 5 mm contact gap and the force share between feet are the only settings chosen on LabValidation: on subjects 2 to 5 against their plates, never against angles.

Forces. Per frame, inverse dynamics gives the generalised forces the movement needs. A residual reduction in the manner of OpenSim's RRA first adjusts the pelvis, trunk and masses within a per-frame cap so that the ground explains the movement. The forces are then split in the two stages of OpenSim's static optimisation [5]: the ground reaction force from the pelvis' six balance equations alone, then the muscle activations with that force fixed. A knee's small free turn is carried by its ligaments, so it no longer splits the vasti. Solved together the problem was ill-posed (a 10^{-6} nudge moved an activation by up to 1.0; in two stages, by at most 5×10^{-3}). What force remains at the pelvis is recorded as the residual, judged against 2% of body weight and 1% of weight times centre-of-mass height [27], and never applied.

One GPU, repeatable. Every stage runs every frame at once on the card with MuJoCo Warp inside Newton, in Warp's [28] deterministic mode: two runs agree bit for bit, which answers, for this chain, the CPU against GPU drift reported on MyoSuite bodies [29].

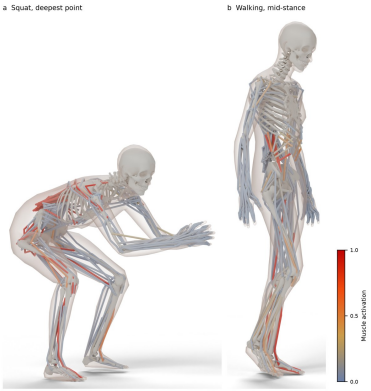


Figure 2: MOTOR's MyoFullBody inside the person's SOMA skin, two LabValidation trials drawn from the run's own saved results: (a) a squat at its deepest point, (b) walking at mid-stance; each muscle coloured by its activation.

3 Validation

LabValidation [1] holds nine healthy subjects filmed by five synchronised iPhones at 60 Hz in walking, walking with trunk sway, squats, sit-to-stand and drop jumps, with marker-based motion capture, force plates and EMG: 144 non-static trials, so 720 single-camera inputs, each run on its own. Angles are scored as in [4] (one time shift per trial; MAE, centred MAE (cMAE), RMSE and r; a per-subject bootstrap of 20,000 draws) on the matched envelope of OpenCap Monocular [2] (the 45° camera, no drop jumps, 15 coordinates). Forces, moments and muscles follow OpenCap [1], the moments against OpenSim's inverse dynamics remade on the lab's own model and settings; activations are compared with the EMG envelopes. Nothing is fitted to the reference angles. Subjects 7 to 11 were never used to choose a setting and are reported apart. "Before" is the previous full run of the same inputs without the contact, ground and ligament changes, scored with the same code.

4 Results

Joint angles. On the matched envelope MOTOR's pooled MAE is 5.16° (95% CI 4.77 to 5.59) over 89 trials of nine subjects, with a centred MAE of 3.06°; on the held-out subjects alone 5.12° (centred 2.91°), and over all 715 inputs 5.35°. For scale, on its own envelopes OpenCap reports 4.5° from two videos [1] and OpenCap Monocular 4.8° [2], and GEM-X's angles taken directly score 4.50° (centred 3.11°) on this envelope [4]. The force changes leave the angles as they were (5.23° before and after on the 72 envelope trials both runs have). Hip flexion keeps the largest error (7.4 and 8.0°, centred 3.5 and 3.8°), and pelvis tilt's 6.1° against a centred 3.6° is an offset.

Ground reaction force. The force under each foot now follows the plates (Figure 3): the median correlation of the vertical force per foot is 0.78 on both feet, from 0.38 and 0.20, and the vertical MAE with both feet summed fell from 53.8 to 16.2% body weight; on the 45° camera without drop jumps it is 11.6%, beside OpenCap's two-camera 11.4% (Table 1). Held-out and training subjects score alike (16.3 and 16.2%). Squats are closest (6.3%), walking and drop jumps furthest (19.6 and 19.3%). The anterior force got worse (6.5 to 7.4%, OpenCap 3.5%).

Joint moments and muscles. Joint moments are not yet at OpenCap's level: 2.41% weight times height against 1.2% (1.6 to 1.9% without drop jumps), and the early-stance knee adduction peak is overestimated (3.5 against 2.1%). The activations follow the EMG better than before (correlation 0.12 to 0.23, best in the vasti at 0.36 to 0.42), and the vasti symmetry now separates squats the EMG labels asymmetric with an AUC of 0.66, from 0.45 (OpenCap 0.83).

Table 1: Kinetics and muscles, all cameras and activities; held out: subjects 7–11. GRF in % body weight (both feet summed), moments in % weight × height; OpenCap: two cameras.

	Before	MOTOR	Held out	OpenCap
Inputs scored	700	715	396	–
GRF vertical MAE	53.8	16.2	16.3	11.4
GRF anterior MAE	6.5	7.4	7.6	3.5
GRF medio-lateral MAE	3.2	3.0	3.0	3.8
Vertical force r per foot (median), right / left	0.38 / 0.20	0.78 / 0.78	0.78 / 0.79	–
Joint moments, OpenCap's 15, MAE	2.45	2.41	2.41	1.2
Early-stance knee adduction peak MAE, left	1.51	1.51	1.67	0.30
Vasti symmetry AUC against the EMG label	0.45	0.66	–	0.83
EMG on/off agreement, mean of 16 channels	0.64	0.66	0.67	–
EMG r, mean of 16 channels	0.12	0.23	0.24	–
Frames within both residual limits	1%	21%	21%	–

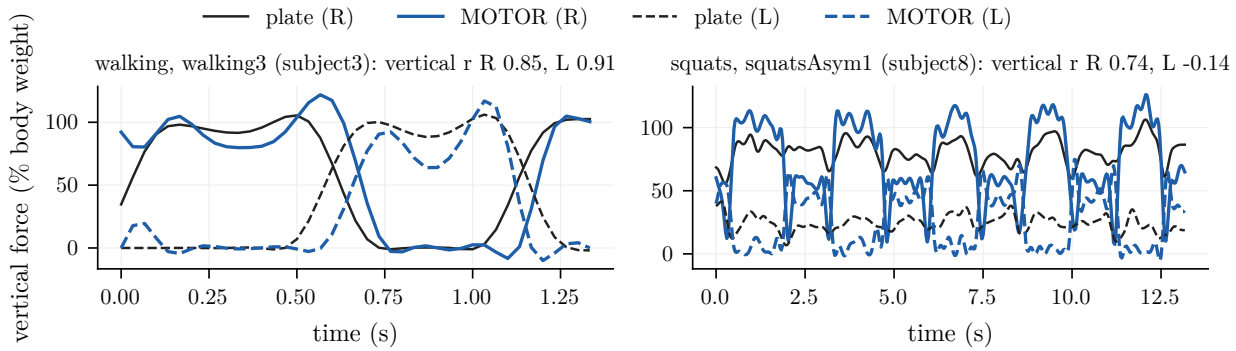


Figure 3: Vertical force per foot against the plates, the 45° camera's median walking and squat trial, low-passed as OpenCap scores them.

Time. From the SOMA body to muscle forces MOTOR takes a median 66 s per trial on one NVIDIA L4 (35 to 789 s), against 2635 to 4196 s for OpenCap's recorded solves on the CPU (Table 9; models and hardware differ). The run took at most 24 cards at once and EUR 29.5.

5 Limitations

Four limits are measured and open. The elbows come out 21 to 24° straighter than the lab's in every envelope trial (centred error 3 to 6°). The offset is GEM-X's arm estimate, not the sim body's: on 540 walking arm-trials GEM-X's own elbow is a median 18.5° straighter than the lab's markers, and the sim body follows it within 2.5°. The pelvis tilt is biased by about 5° (one direction in 82 of 89 envelope trials). The pelvis residual stays above OpenSim's limits (2% body weight, 1% weight times centre-of-mass height; medians 7.0% and 8.9%), so the forces are not yet dynamically consistent with the movement. The activation pattern is still partly off: rectus femoris does not follow its EMG ($r = -0.08$) and gluteus medius agrees on or off on only half the frames. LabValidation is one small healthy cohort.

6 Conclusion

MOTOR gives foot forces, moments and 416 muscle forces from one video in a minute per trial on one GPU.

References

[1] Scott D. Uhlrich, Antoine Falisse, Łukasz Kidziński, Julie Muccini, Michael Ko, Akshay S. Chaudhari, Jennifer L. Hicks, and Scott L. Delp. OpenCap: Human movement dynamics from smartphone videos. *PLOS Computational Biology*, 19(10):e1011462, 2023.

[2] Selim Gilon, Emily Y. Miller, and Scott D. Uhlrich. OpenCap Monocular: 3D human kinematics and musculoskeletal dynamics from a single smartphone video. *arXiv preprint arXiv:2603.24733*, 2026.

[3] J. D. Peiffer, Kunal Shah, Irina Djuraskovic, Shawana Anarwala, Kayan Abdou, Rujvee Patel, Prakash Jayabalan, Brenton Pennicooke, and R. James Cotton. Portable biomechanics laboratory: Clinically accessible movement analysis from a handheld smartphone. *arXiv preprint arXiv:2507.08268*, 2025.

[4] Joey Páolo Kardolus, Daan Hendriks, and Jaap Jansen. Direct clinical joint angle extraction from parametric body model rotation matrices. *arXiv preprint arXiv:2607.17639*, 2026.

[5] Scott L. Delp, Frank C. Anderson, Allison S. Arnold, Peter Loan, Ayman Habib, Chand T. John, Eran Guendelman, and Darryl G. Thelen. OpenSim: Open-source software to create and analyze dynamic simulations of movement. *IEEE Transactions on Biomedical Engineering*, 54 (11):1940–1950, 2007.

[6] Apoorva Rajagopal, Christopher L. Dembia, Matthew S. DeMers, Denny D. Delp, Jennifer L. Hicks, and Scott L. Delp. Full-body musculoskeletal model for muscle-driven simulation of human gait. *IEEE Transactions on Biomedical Engineering*, 63(10):2068–2079, 2016. doi: 10.1109/TBME.2016.2586891.

[7] Christopher L. Dembia, Nicholas A. Bianco, Antoine Falisse, Jennifer L. Hicks, and Scott L. Delp. OpenSim Moco: Musculoskeletal optimal control. *PLOS Computational Biology*, 16(12): e1008493, 2020. doi: 10.1371/journal.pcbi.1008493.

[8] R. James Cotton. Differentiable biomechanics unlocks opportunities for markerless motion capture. *arXiv preprint arXiv:2402.17192*, 2024.

[9] R. James Cotton. KinTwin: Imitation learning with torque and muscle driven biomechanical models enables precise replication of able-bodied and impaired movement from markerless motion capture. *arXiv preprint arXiv:2505.13436*, 2025.

[10] R. James Cotton, J. D. Peiffer, Lucinda Williamson, John Leske, and Georgios Pavlakos. Biomechanical 3D body: Self-supervised distillation of biomechanical pose from a 3D body foundation model. *arXiv preprint arXiv:2608.29928*, 2026.

[11] Vittorio Caggiano, Huawei Wang, Guillaume Durandau, Massimo Sartori, and Vikash Kumar. MyoSuite: A contact-rich simulation suite for musculoskeletal motor control. *arXiv preprint arXiv:2205.13600*, 2022.

[12] Firas Al-Hafez, Guoping Zhao, Jan Peters, and Davide Tateo. LocoMuJoCo: A comprehensive imitation learning benchmark for locomotion. *arXiv preprint arXiv:2311.02496*, 2023.

[13] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012.

[14] Chengkun Li, Cheryl Wang, Bianca Ziliotto, Merkourios Simos, Jozsef Kovecses, Guillaume Durandau, and Alexander Mathis. Towards embodied AI with MuscleMimic: Unlocking full-body musculoskeletal motor learning at scale. *arXiv preprint arXiv:2603.25544*, 2026.

[15] Yunyue Wei, Chenhui Zuo, Shanning Zhuang, Haixin Gong, Yaming Liu, and Yanan Sui. Scaling whole-body human musculoskeletal behavior emulation for specificity and diversity. *arXiv preprint arXiv:2603.29332*, 2026.

[16] Jenishkumar Chauhan, Samarth Brahmbhatt, and Vineet Vashista. Towards real-world identification of fatigued muscle groups via musculoskeletal simulation. *arXiv preprint arXiv:2605.26151*, 2026.

[17] Sagar Verma. emg2tendon: From sEMG signals to tendon control in musculoskeletal hands. *arXiv preprint arXiv:2508.08269*, 2025.

[18] The Newton Developers. Newton: a GPU-accelerated physics simulation engine, 2026. URL https://github.com/newton-physics/newton. Version 1.6.1 used here (SolverMuJoCo). Apache-2.0.

[19] Google DeepMind and NVIDIA. MuJoCo Warp: a GPU-optimized version of the MuJoCo physics simulator, 2026. URL https://github.com/google-deepmind/mujoco_warp. Apache-2.0.

[20] NVIDIA. GEM-X: Monocular whole-body 3D human pose estimation using the SOMA body model, 2026. URL https://github.com/NVlabs/GEM-X. Apache-2.0.

[21] Jiefeng Li, Jinkun Cao, Haotian Zhang, Davis Rempe, Jan Kautz, Umar Iqbal, and Ye Yuan. GENMO: A GENeralist model for human MOtion. *arXiv preprint arXiv:2505.01425*, 2025.

[22] Jun Saito, Jiefeng Li, Michael de Ruyter, Miguel Guerrero, et al. SOMA: Unifying parametric human body models. *arXiv preprint arXiv:2603.16858*, 2026.

[23] Ruicheng Wang et al. MoGe-2: Accurate monocular geometry with metric scale and sharp details. *arXiv preprint arXiv:2507.02546*, 2025.

[24] Katherine R. S. Holzbaur, Wendy M. Murray, and Scott L. Delp. A model of the upper extremity for simulating musculoskeletal surgery and analyzing neuromuscular control. *Annals of Biomedical Engineering*, 33(6):829–840, 2005. doi: 10.1007/s10439-005-3320-7.

[25] Katherine R. Saul, Xiao Hu, Craig M. Goehler, Meghan E. Vidt, Melissa Daly, Anca Velisar, and Wendy M. Murray. Benchmarking of dynamic simulation predictions in two software platforms using an upper limb musculoskeletal model. *Computer Methods in Biomechanics and Biomedical Engineering*, 18(13):1445–1458, 2015. doi: 10.1080/10255842.2014.916698.

[26] Kevin Zakka. mink: Python inverse kinematics based on MuJoCo, 2024. URL https://github.com/kevinzakka/mink. Apache-2.0.

[27] Jennifer L. Hicks, Thomas K. Uchida, Ajay Seth, Apoorva Rajagopal, and Scott L. Delp. Is my model good enough? best practices for verification and validation of musculoskeletal models and simulations of movement. *Journal of Biomechanical Engineering*, 137(2):020905, 2015. doi: 10.1115/1.4029304.

[28] Miles Macklin. Warp: A high-performance Python framework for GPU simulation and graphics, 2022. URL https://github.com/NVIDIA/warp. Version 1.18.0 used here. Apache-2.0 since 1.6.2.

[29] M. Colley. Direct MyoSuite CPU/CUDA muscle rollouts drift at the wrist with contacts disabled. MuJoCo Warp issue 1835, 2026. URL https://github.com/google-deepmind/mujoco_warp/issues/1835.

A Full technical validation

Laid out as Appendix B of [4]. All 720 single-camera inputs (every activity including drop jumps, each camera its own input) were prepared and sent to the cluster; 715 ran and were scored, and Appendix B gives the rest by cause. "Before" is the previous full run of the same 720 inputs (700 scored) with the chain as it was before the contact, ground and ligament changes, scored again with this run's code. Every number is from paper/evidence/sim-body/labval-final-2026-11-07.txt.

A.1 Every joint angle

Table 2: MAE and centred MAE (degrees) per coordinate, previous full run → this run, and this run's mean r (†: the reference moves less than 15°). Envelope: 45° camera, no drop jumps (72 → 89 trials). All: every scored input (700 → 715). The matched 15 first; pro_sup has no shared zero, so only its cMAE.

	Envelope			All		
Coordinate	MAE	cMAE	r	MAE	cMAE	r
ankle angle r	6.41 → 6.39	3.18 → 3.19	0.89	5.97 → 5.96	3.94 → 3.92	0.92
ankle angle l	4.45 → 4.61	3.22 → 3.32	0.89	5.52 → 5.51	3.75 → 3.75	0.92
knee angle r	6.30 → 6.06	3.16 → 3.04	0.99	6.02 → 5.99	3.18 → 3.17	0.99
knee angle l	6.10 → 5.94	3.33 → 3.24	0.98	6.11 → 6.09	3.51 → 3.50	0.98
hip flexion r	8.17 → 8.03	3.90 → 3.82	0.98	8.91 → 8.89	4.16 → 4.11	0.98
hip flexion l	7.46 → 7.37	3.56 → 3.53	0.98	8.96 → 8.93	4.22 → 4.17	0.98
hip adduction r	4.20 → 4.20	2.81 → 2.73	0.52†	3.69 → 3.72	2.58 → 2.59	0.52†
hip adduction l	4.78 → 4.72	2.83 → 2.76	0.60†	3.48 → 3.52	2.50 → 2.52	0.57†
hip rotation r	4.10 → 4.07	2.62 → 2.62	0.43†	4.38 → 4.36	2.77 → 2.77	0.51
hip rotation l	4.53 → 4.41	3.19 → 3.06	0.43†	4.22 → 4.20	3.03 → 3.00	0.54
pelvis tilt	5.93 → 6.06	3.60 → 3.64	0.51	7.05 → 7.02	4.43 → 4.39	0.40
pelvis list	3.26 → 3.15	2.48 → 2.46	0.17†	3.10 → 3.11	2.22 → 2.24	0.16†
lumbar extension	4.66 → 4.56	3.15 → 3.11	0.64	5.36 → 5.32	3.37 → 3.35	0.75
lumbar bending	4.37 → 4.19	3.31 → 3.21	0.44	3.68 → 3.69	2.51 → 2.54	0.45†
lumbar rotation	3.67 → 3.67	2.14 → 2.12	0.58†	4.01 → 3.97	2.74 → 2.71	0.44
arm add l	3.79 → 3.98	2.33 → 2.34	0.70	6.62 → 6.58	5.18 → 5.12	0.70
arm add r	4.30 → 4.23	2.25 → 2.23	0.51†	6.22 → 6.16	4.45 → 4.40	0.64
arm flex l	7.92 → 8.03	3.11 → 3.19	0.93	12.59 → 12.46	8.72 → 8.59	0.90
arm flex r	6.77 → 7.00	3.09 → 3.19	0.93	11.32 → 11.20	8.65 → 8.52	0.90
arm rot l	8.19 → 8.01	4.41 → 4.50	0.64	12.48 → 12.35	9.91 → 9.78	0.67
arm rot r	7.25 → 7.03	4.27 → 4.09	0.67	11.31 → 11.18	8.74 → 8.61	0.68
elbow flex l	23.80 → 23.68	4.74 → 4.61	0.74	24.77 → 24.70	5.73 → 5.67	0.79
elbow flex r	20.96 → 21.12	3.07 → 3.00	0.80	22.79 → 22.74	5.84 → 5.76	0.76
mtp angle l	15.23 → 15.35	2.49 → 2.47	0.15†	15.73 → 15.74	2.19 → 2.19	0.10†
mtp angle r	15.17 → 15.22	2.58 → 2.55	-0.02†	14.99 → 14.99	2.05 → 2.06	0.00†
pelvis rotation	2.76 → 2.70	2.63 → 2.57	0.50†	2.66 → 2.63	2.45 → 2.42	0.48†
pro sup l	-	2.82 → 2.82	0.16†	-	3.63 → 3.61	0.25
pro sup r	-	2.44 → 2.55	0.18†	-	3.48 → 3.46	0.26†
subtalar angle l	9.18 → 9.24	7.02 → 6.88	-0.21	9.74 → 9.73	6.25 → 6.25	-0.07
subtalar angle r	9.02 → 9.16	6.26 → 6.26	0.04	11.44 → 11.39	6.21 → 6.21	0.04
matched 15, pooled	5.23 → 5.16	3.10 → 3.06		5.36 → 5.35	3.26 → 3.25	

The angles are the same chain's in both runs (GEM-X and the fit are unchanged); the small differences come from the inputs each run has. The elbows are about 21 to 25° off with a centred error of 3 to 6° (straighter than the lab's in each of the 89 envelope trials; GEM-X's arm estimate,

which the sim body follows), and the toes (mtp) 15° off with a centred 2 to 3°: constant offsets, not tracking errors.

A.2 Per subject group, activity and camera

Table 3: This run per group (the previous full run's value in brackets): inputs, matched MAE and cMAE (degrees), vertical GRF MAE (% body weight, both feet), median vertical r per foot, joint moment MAE over OpenCap's 15 (% weight × height), mean EMG r over the channels. Cameras: LabValidation's numbering; Cam1 is the envelope's 45° camera.

Group	n	MAE	cMAE	GRF vert.	r R / L	Moments	EMG r
held out 7–11	396	5.13 (5.14)	3.02 (3.02)	16.28 (54.33)	0.77 / 0.79	2.41 (2.43)	0.24
training 2–5	319	5.62 (5.65)	3.54 (3.56)	16.15 (53.07)	0.78 / 0.77	2.42 (2.48)	0.22
walking	135	4.38 (4.40)	2.92 (2.93)	19.55 (76.11)	0.82 / 0.80	1.88 (1.79)	0.27
walking (trunk sway)	135	4.67 (4.67)	3.05 (3.05)	16.31 (84.14)	0.82 / 0.84	1.72 (1.53)	0.27
squats	87	5.79 (5.79)	3.26 (3.28)	6.32 (47.18)	0.53 / 0.47	1.58 (1.54)	0.17
sit-to-stand	90	5.39 (5.34)	2.92 (2.91)	11.58 (47.71)	0.68 / 0.80	1.55 (1.74)	0.20
drop jump	268	6.03 (6.03)	3.63 (3.63)	19.27 (32.21)	0.78 / 0.78	3.59 (3.73)	0.22
Cam0	144	5.21 (5.20)	3.14 (3.14)	14.95 (54.10)	0.75 / 0.83	2.44 (2.44)	0.23
Cam1 (45°)	141	5.62 (5.72)	3.33 (3.40)	15.50 (53.27)	0.73 / 0.81	2.54 (2.65)	0.22
Cam2	144	5.36 (5.36)	3.53 (3.53)	15.54 (52.72)	0.81 / 0.82	2.21 (2.31)	0.25
Cam3	143	5.35 (5.35)	3.20 (3.21)	15.96 (53.03)	0.81 / 0.70	2.44 (2.44)	0.23
Cam4	143	5.23 (5.23)	3.05 (3.05)	19.16 (55.68)	0.78 / 0.75	2.43 (2.45)	0.23

A.3 Ground reaction force

Table 4: Ground reaction force per activity, this run (previous run in brackets): MAE in % body weight on three axes with both feet summed, as OpenCap scored it; the share of the plate's stance frames the run's foot carries.

Activity	vertical	anterior	medio-lateral	3-axis mean	stance carried R / L
all	16.22 (53.77)	7.41 (6.45)	2.99 (3.19)	8.9 (21.1)	0.82 / 0.79
walking	19.55 (76.11)	12.03 (9.65)	3.02 (3.35)	11.5 (29.7)	0.82 / 0.75
walking (trunk sway)	16.31 (84.14)	10.86 (9.11)	3.49 (4.21)	10.2 (32.5)	0.83 / 0.85
squats	6.32 (47.18)	3.53 (2.56)	3.30 (4.17)	4.4 (18.0)	0.99 / 0.95
sit-to-stand	11.58 (47.71)	4.42 (4.74)	3.25 (3.51)	6.4 (18.7)	0.82 / 0.82
drop jump	19.27 (32.21)	5.61 (5.37)	2.54 (2.20)	9.1 (13.3)	0.76 / 0.71
OpenCap, 2 cameras	11.4	3.5	3.8	6.2	–

A.4 Residuals

Table 5: The pelvis residual this run leaves, per group: the median over trials of each trial's median force (% body weight) and moment (% weight × centre-of-mass height), the median of each trial's 95th-percentile force, and the mean share of frames within both of OpenSim's limits (2% and 1%). No trial is within them on every frame.

Group	trials	force	moment	force, 95th pct.	frames within
all	715	7.0	8.90	24.4	21%
held out 7–11	396	7.1	9.31	23.9	21%
walking	135	7.8	10.99	23.0	6%
walking (trunk sway)	135	7.0	9.59	20.3	7%
squats	87	0.7	1.06	8.4	49%
sit-to-stand	90	0.2	2.82	8.8	46%
drop jump	268	10.9	9.32	85.5	17%

A.5 Muscle activations against EMG

Table 6: Each EMG channel against MOTOR's activation of that muscle, all 715 inputs: the share of frames on which both are on or both off, Pearson's r, and the lag (s) at the cross-correlation peak.

Muscle	on/off	r	lag	Muscle	on/off	r	lag
bflh l	0.65	0.13	-0.130	semiten r	0.70	0.23	-0.034
bflh r	0.65	0.13	-0.144	soleus l	0.70	0.26	-0.030
gasmed l	0.70	0.23	-0.009	soleus r	0.69	0.28	-0.041
gasmed r	0.69	0.26	-0.022	tibant l	0.59	0.22	+0.031
glmed1 l	0.51	0.13	+0.051	vaslat l	0.72	0.40	-0.003
glmed1 r	0.52	0.15	+0.040	vaslat r	0.72	0.42	-0.018
recfem l	0.56	-0.08	-0.002	vasmed l	0.72	0.36	-0.024
semiten l	0.66	0.18	-0.053	vasmed r	0.76	0.41	-0.033

Table 7: OpenCap's vasti symmetry index in squats: the AUC of MOTOR's index against the asymmetry the instructions asked for and against the EMG's own label (index over 1.15), previous run → this run. OpenCap: 0.83 against the EMG label.

Camera	AUC, instructed	AUC, EMG label
Cam0	0.47 → 0.84	0.64 → 0.75
Cam1	0.52 → 0.72	0.08 → 0.62
Cam2	0.60 → 0.95	0.52 → 0.84
Cam3	0.75 → 0.90	0.65 → 0.83
Cam4	0.53 → 0.81	0.25 → 0.48
all	0.56 → 0.73	0.45 → 0.66

A.6 Time per step

Table 8: MOTOR's sim body wall seconds per step on one NVIDIA L4, over the 715 runs: median, range, and the median per frame (clips of 30 Hz).

Step	median s	min s	max s	s per frame
fit	21.825	14.483	623.317	0.303
set up: Newton's model, its card, the filtered path	4.697	3.878	32.851	0.061
footing	4.669	1.758	35.312	0.062
RRA, rounds	19.862	5.910	392.306	0.246
RRA, follows and frame caps	4.603	1.714	45.888	0.061
checks: RRA's record, feet held, following on MuJoCo's compile	2.120	1.111	17.879	0.028
static optimisation 1, ground reaction	3.003	0.114	28.318	0.034
static optimisation 2, stops and muscles	2.296	1.169	22.356	0.032
inverse dynamics, contacts and muscle paths	1.120	0.356	12.459	0.013
records and push measures	0.264	0.042	2.639	0.003
arrays	0.005	0.003	0.012	0.000
total	66.214	34.843	788.668	0.871

Table 9: Wall time per trial. OpenCap's solves are from its LabValidation release (80 muscles, windows of 0.99 to 1.68 s, solve only, hardware not recorded). Figure 4 draws them on one scale.

Chain	Clip	Wall time
OpenCap solve [1]	walk, drop jump, squat	4196, 4096, 2635 s
Moco, MocoTrack [7]	gait	about 130 min
OpenSim CMC [6]	gait	about 10 min
MOTOR sim body, one L4	LabValidation, 715 trials	66 s median (35–789 s)

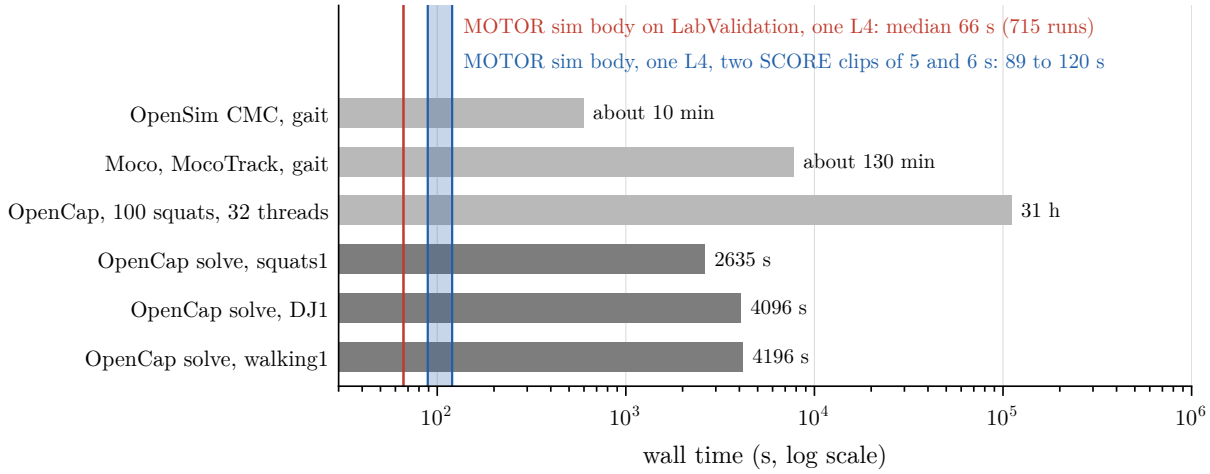


Figure 4: Wall time per trial, log scale: OpenCap's recorded solves and published OpenSim times (grey), MOTOR's sim body on two SCORE clips (blue band) and its median on LabValidation (red).

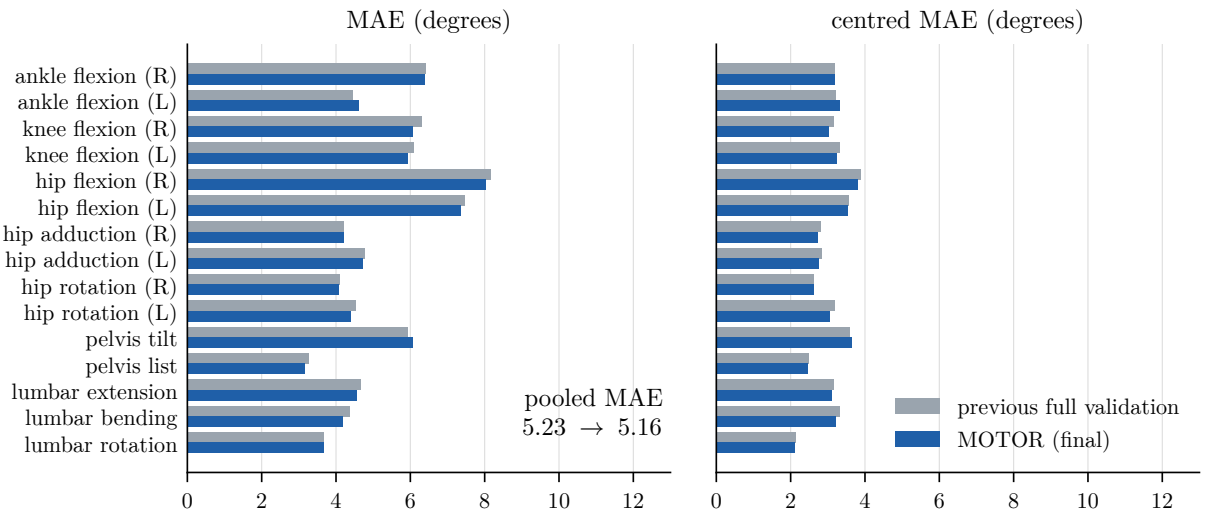


Figure 5: Per-coordinate MAE and centred MAE on the matched envelope, previous full run and this run.

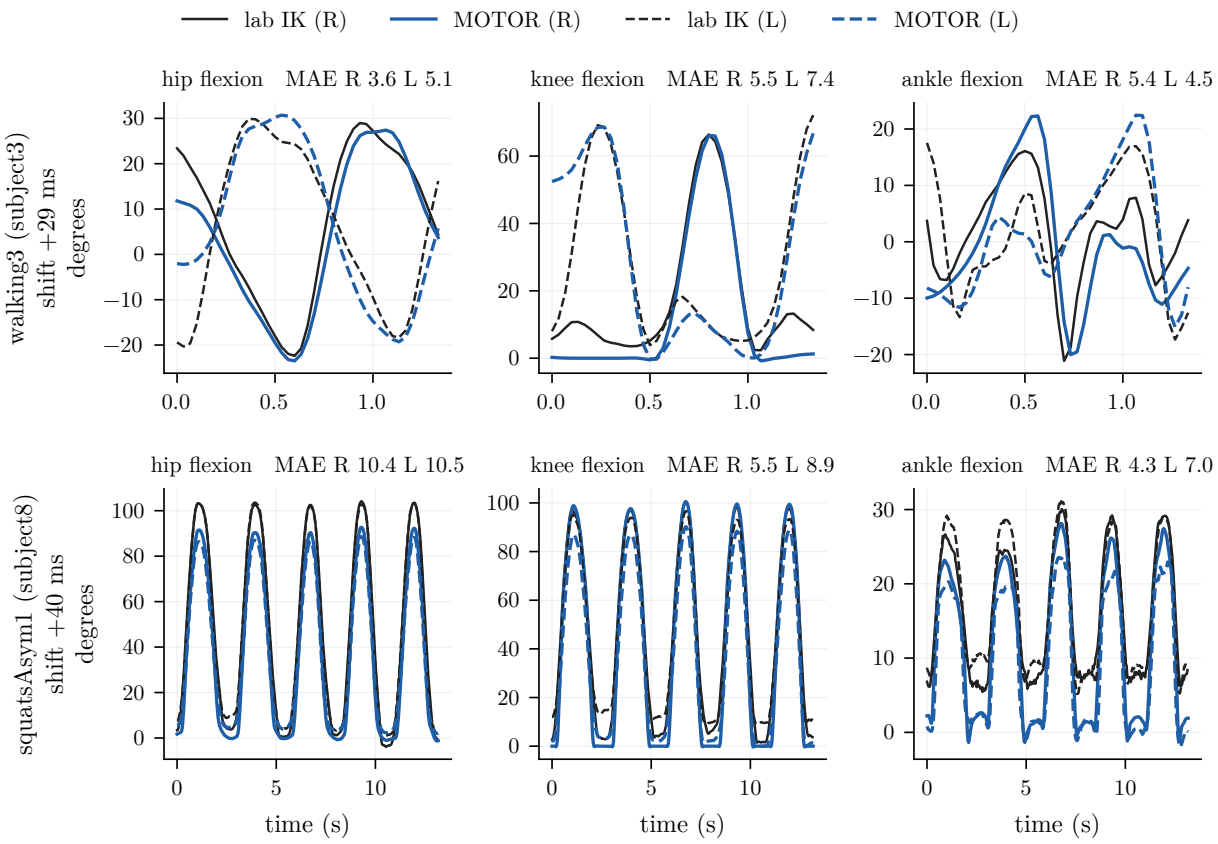


Figure 6: Hip, knee and ankle flexion of the 45° camera's median walking and squat trial (by angle error), MOTOR against the lab's inverse kinematics moved by the scored time shift.

B Failures

Table 10: The 5 of 720 inputs without a score in this run, by cause.

Cause	Inputs
The card's fit solve did not stop in 3000 steps (subject 10, squats1, two cameras)	2
A frame's fit did not converge in 300 steps (two drop jumps)	2
Passed its job's time limit (subject 10, squats1, a third camera)	1
Total	5

The run first stopped at its EUR 25 spend limit with 3 of its 48 jobs unfinished; their 45 inputs ran after with the same code.

Two faults found in the first scoring of an earlier run are fixed and reported because they moved the headline number then. First, 148 of the 720 GEM-X results came from another video: a resumed cloud run named its jobs by position in a shortened list, so a job picked up an earlier job's finished output; jobs are now named by a hash of their own description. Second, the time shift could not reach a walking trial's true lag: requiring the shifted reference to cover 80% of a clip about one gait cycle long capped the shift near ± 0.27 s, against the protocol's ± 1.0 s; at 50% the feet are in phase at the found lag in every walking trial. The pooled matched MAE went from 8.59° (78 trials) to 6.02° (62 trials) then.

C Disclosure

The code of MOTOR and of its validation was written by a coding agent (Claude Code) under the author's direction and review; this draft was written the same way. No result was hand-picked. Three contact settings and the box clearance were chosen on LabValidation's subjects 2 to 5 against their force plates, never against joint angles; subjects 7 to 11 were held out and are reported apart. The figures' trials were picked by rule (the median trial by angle error), not by eye.